Shruti Sriram

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EDUCATION

The University of Texas at Austin — *Aug 2024 - May 2026*
Master of Science in Computer Science — *CGPA: 3.85/4.0*
Focus Areas: *Software Development, AI/ML, Computer Vision*
Courses: *Visual Recognition, Generative Models in ML, Spoken Language Technologies, Statistics, Database Systems, Distributed Computing*

Sri Sivasubramaniya Nadar College of Engineering — *Aug 2020 - Jun 2024*
Bachelor of Engineering in Computer Science and Engineering (Gold Medal in Semester 1, Silver Medal in Semester 3) — *CGPA: 9.61/10*

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, HTML, CSS, JavaScript, Bash, Go, Rust
Tools & Frameworks: React, Node.js, Tableau, Apache Camel, Spring Boot
ML Stack: PyTorch, TensorFlow, Keras, OpenCV, Scikit-Learn, Numpy, Pandas
AI Frameworks and Agentic Systems: OpenAI API, Claude, Cursor, Ollama, LangChain, CrewAI
Databases: MySQL, PostGreSQL, DuckDB, MongoDB
Cloud & DevOps: AWS, Git

EXPERIENCE

The University of Texas at Austin, Graduate Researcher (System Level Design Group) — *Oct 2025 – Present*
- Designed a novel, computationally efficient medical image segmentation architecture for cross-domain settings, improving DICE by up to +18.9% on challenging domain-shifted datasets
- Developed an edge-optimized deployment pipeline for the model to reduce inference latency and memory footprint on resource-constrained devices
- Built a real time segmentation device for colorectal procedures, enabling high accuracy with low computation to generate instantaneous segmentation masks
- Developing an end-to-end image segmentation pipeline with a patch-selective super resolution module (trained using curriculum learning) for cross-domain generalization
- Developing a pipeline to diagnose the type of uveitis (retinal abnormality), and predict and visualize disease progression to aid ophthalmologists

Ascendion, Inc., Engineering Intern — *Jun 2025 – August 2025*
- Built code analysis tools for the AAVA Edge platform, including Abstract Syntax Tree (AST) parsers (using pycparser, javalang, and esprima), visualizations, and call graphs, to support faster code understanding and refactoring
- Developed a DuckDB backend and Thespian actor abstraction to store parsed ASTs and AI agent outputs; this set-up enables multi-user access and versioning, improves data accessibility and reliability, thus enhancing code traceability and reproducibility

Citibank, Summer Analyst Intern — *May 2023 – Jul 2023*
- Re-engineered legacy Ab Initio billing feed pipeline to automate periodic ingestion of 50K+ billing records per day using Apache Camel, Java, and Spring Boot
- Reduced manual intervention by 90% by scheduling recurring jobs using cron expressions
- Built a Tableau dashboard on geo-temporal data to help business analysts monitor trends, spot anomalies, and support forecasting and planning
- Performed hypothesis testing to identify temporal anomalies and helped stakeholders frame data-driven questions more effectively

Synergy Maritime, Software Development Intern — *Aug 2022 - Aug 2022*
- Designed an internal web app to manage ship details, including IMO numbers, vessel dimensions, and voyage records
- Developed a frontend using HTML/CSS and SQL Server–backed queries with integrated form validation, to optimize data retrieval time from 15 seconds to under 5 seconds
- Built dashboards using Pandas, NumPy, and Matplotlib to visualize 100K+ records from vessel sensors and operations, and aid anomaly detection

PUBLICATIONS

SACNet: Scale-Adaptive Convolutional Decoder for Generalizable Polyp Segmentation — Submitted to WACV 2027
Multimodal Forgery Detection in Videos Using a Tri-Network Model *(Paper Link)* — IEEE Xplore
PoSh at SemEval-2023 Task 10: Explainable Detection of Online Sexism *(Paper Link)* — ACL Anthology
A Fusion Approach for Web Search Result Diversification Using Machine Learning Algorithms *(Paper Link)* — CEUR Workshop

PROJECTS

Review-Based Business Recommendation System — *Jul 2026 – Aug 2026*
- Built and deployed a multi-agent AI recommendation platform using LangChain, Amazon Bedrock, Titan Embeddings, Pinecone, and AWS ECS to retrieve, evaluate, and summarize live Google Maps reviews with confidence-aware rankings.

LLM Prompt Engineering for Data Cleaning (PHI-3) — *Aug 2024 – Dec 2024*
- Identified optimal prompt strategies by evaluating Phi-3's effectiveness in cleaning files with diverse error types, by benchmarking multiple prompt patterns to measure output quality

911 Emergency Response System Enhancement — *Nov 2024 – Nov 2024*
- Built a multi-agent AI system to augment 911 emergency dispatch, with specialized agents for threat analysis, multilingual translation, and multimodal input routing
- Each agent processes a distinct signal: caller tone, background audio, or video input, using LLaMA 3.2-Vision
- Improved dispatcher prioritization and accessibility for non-English speakers and disabled users

Segmentation and 3D Reconstruction of Arteries, Istinye University, Turkey — *Dec 2022 – Oct 2024*
- Developed a pipeline for arterial segmentation and 3D reconstruction using UNet and TransUNet, integrating K-means clustering and edge detection to generate precise geometric models
- Achieved significantly higher segmentation fidelity with UNet (IoU 0.89 vs. 0.65), enabling more reliable anatomical modeling for downstream clinical analysis and surgical planning

Automatic Estimation of Fish Count in Aqua Farms — *Jan 2022 – May 2024*
- Built a YOLOv8- and ByteTrack-based fish counting and tracking system, using a Raspberry Pi and a webcam, for smart aquaculture and automated feed-dispensing